Universidad del Desarrollo

Class 4: Introduction to Machine Learning

Unit II: **Data Preparation and Predictive Models**

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Shifting the Analytics Paradigm: Descriptive - Predictive BI

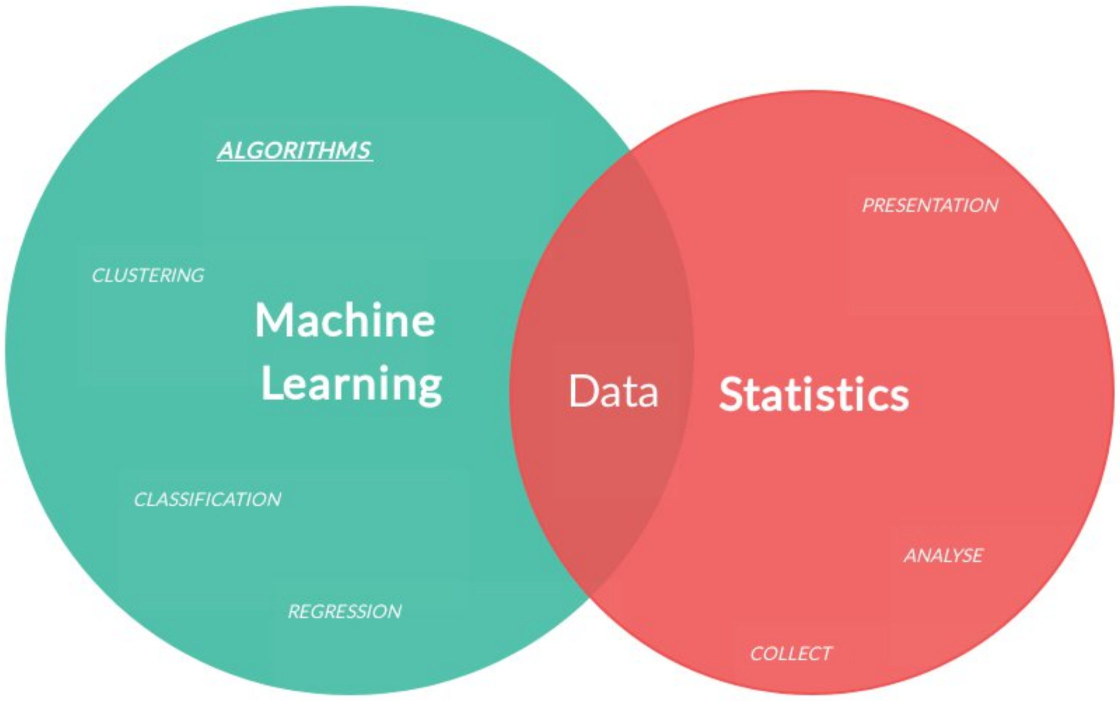
The business goal

- **Transitioning from Insight to Action:** Moving beyond traditional reporting by treating historical data as the raw material for predictive models.

- **Quantifying Uncertainty:** Shifting the decision-making process from intuition toward statistically validated frameworks that measure risk and expected outcomes.

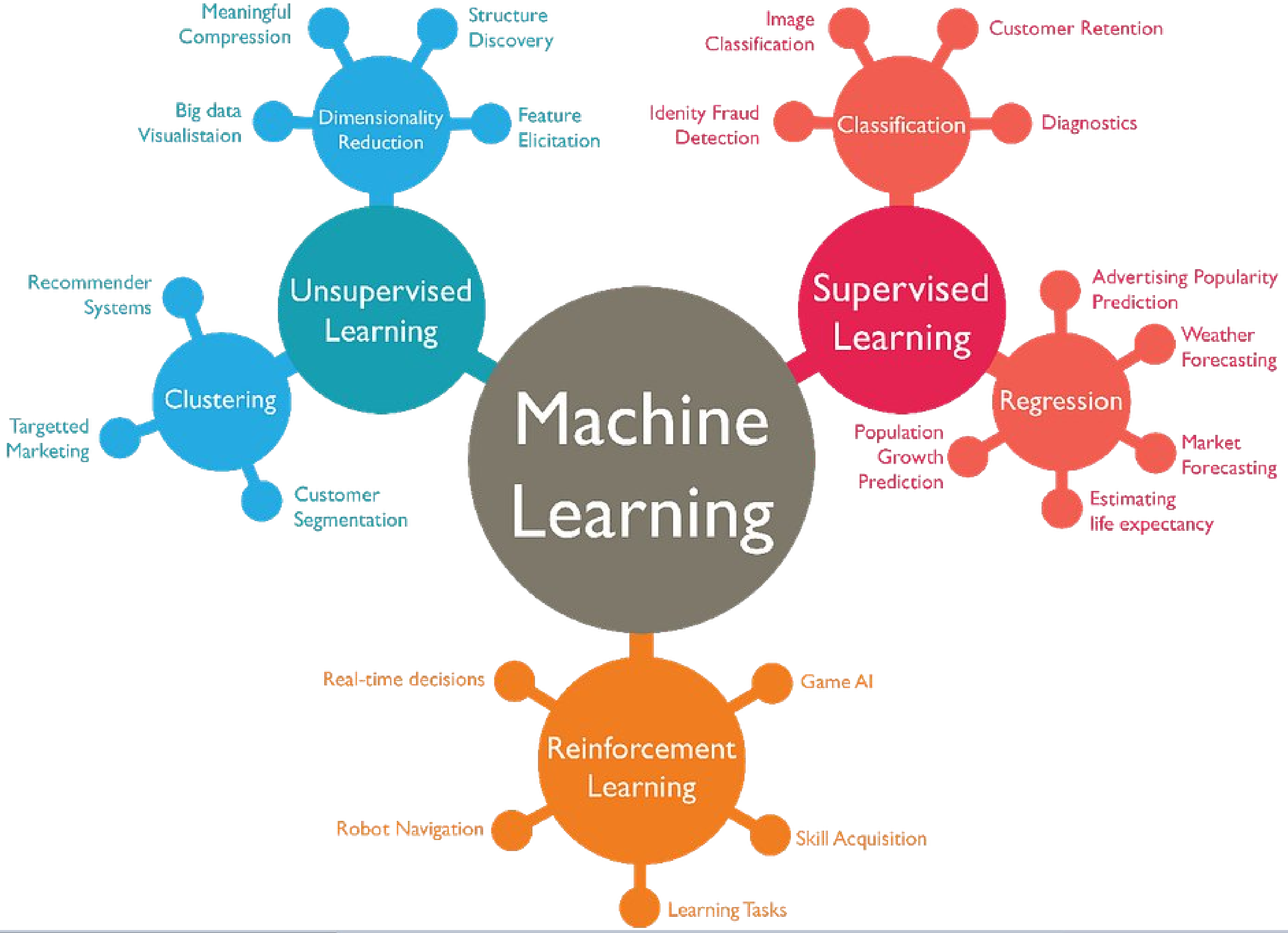
- **Operational Integration:** Embedding predictive capabilities into core business processes to optimize resource allocation and cost efficiency.

- **Driving Strategic Value:** Creating a competitive advantage by anticipating market shifts and customer behavior before they impact the bottom line.



Machine Learning Taxonomy

Architectural Selection Based on Data Structures



Machine Learning Taxonomy

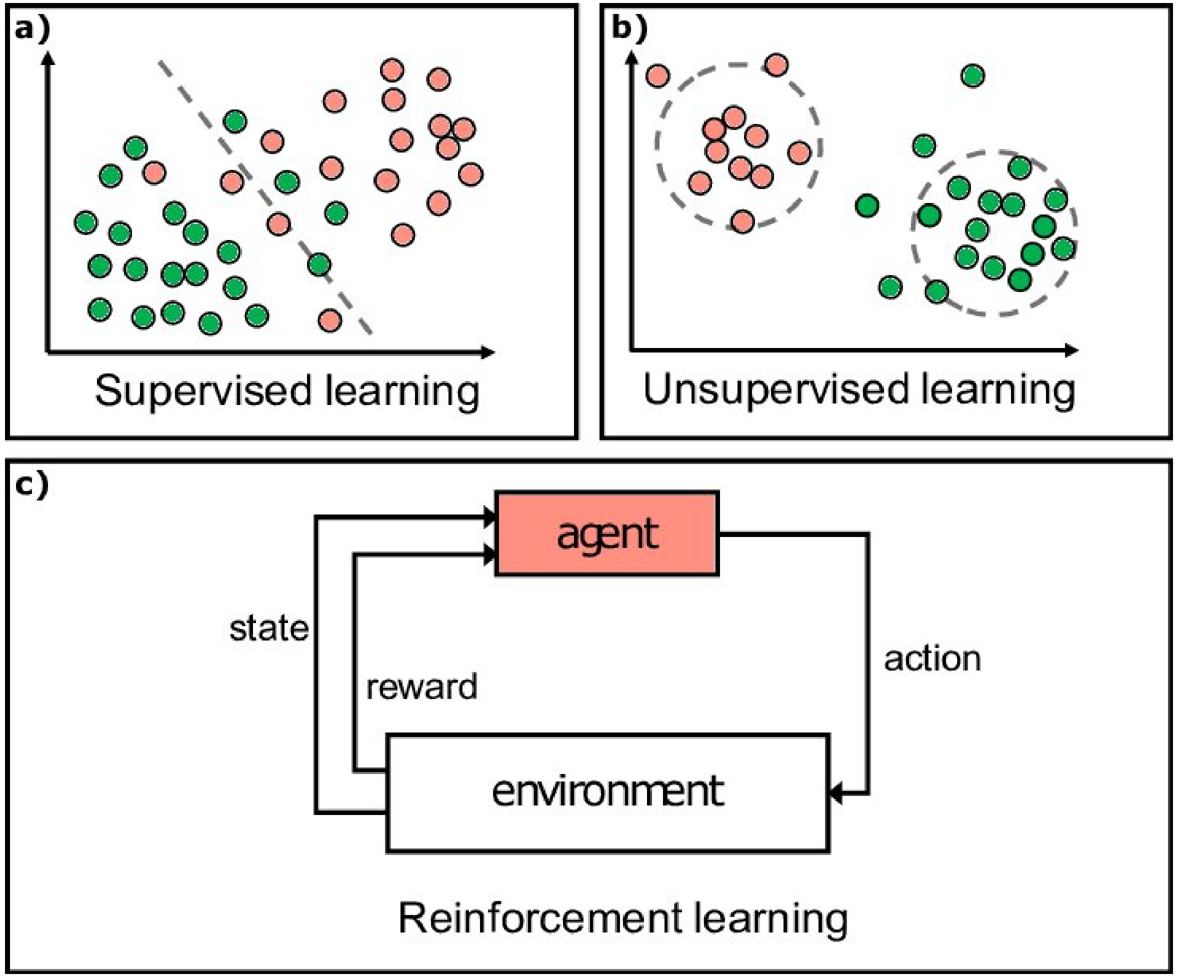
Architectural Selection Based on Data Structures

- **Signal-Based Classification:** Categorizing models by the type of feedback available—shifting from predictive mapping to structural discovery.

- **Supervised Learning Architectures:** Establishing functional relationships between input features and target variables for regression or classification.

- **Unsupervised Pattern Extraction:** Identifying latent structures and natural clusters in high-dimensional data without predefined output labels.

- **Reinforcement Learning & Dynamic Control:** Optimizing sequential decision-making through agent-environment interaction to maximize long-term performance (rewarding)



Supervised Learning - Regression vs Classification

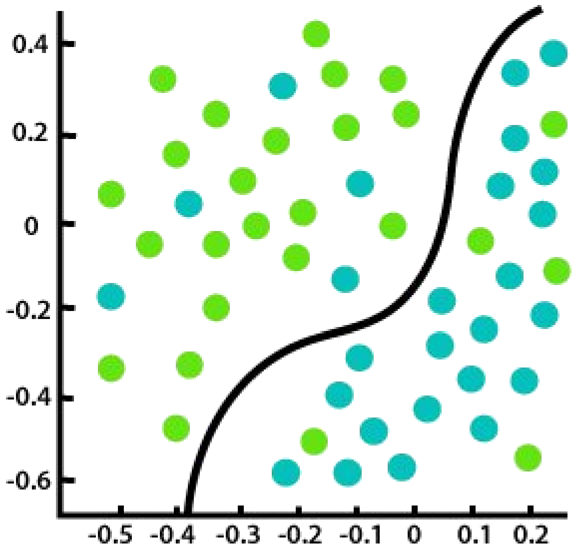
Categorizing Supervised Learning Objectives

- **Output Dimensionality:** Distinguishing between continuous numeric variables and discrete categorical boundaries to define the model architecture.

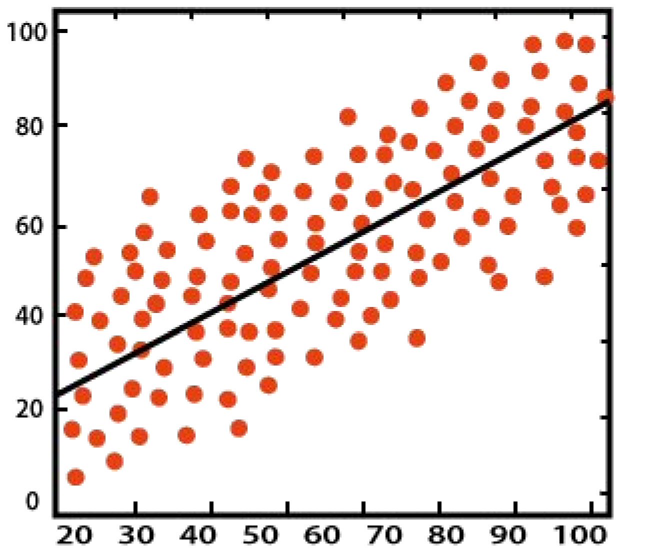
- **Regression Frameworks:** Modeling the conditional expectation of a quantitative target, such as estimating asset valuation or demand volume.

- **Classification Logic:** Partitioning the feature space into mutually exclusive classes to predict event probability or system states.

- **Impact on Decision Metrics:** Aligning the output nature with specific KPIs—shifting from error variance in regression to accuracy and precision in classification.



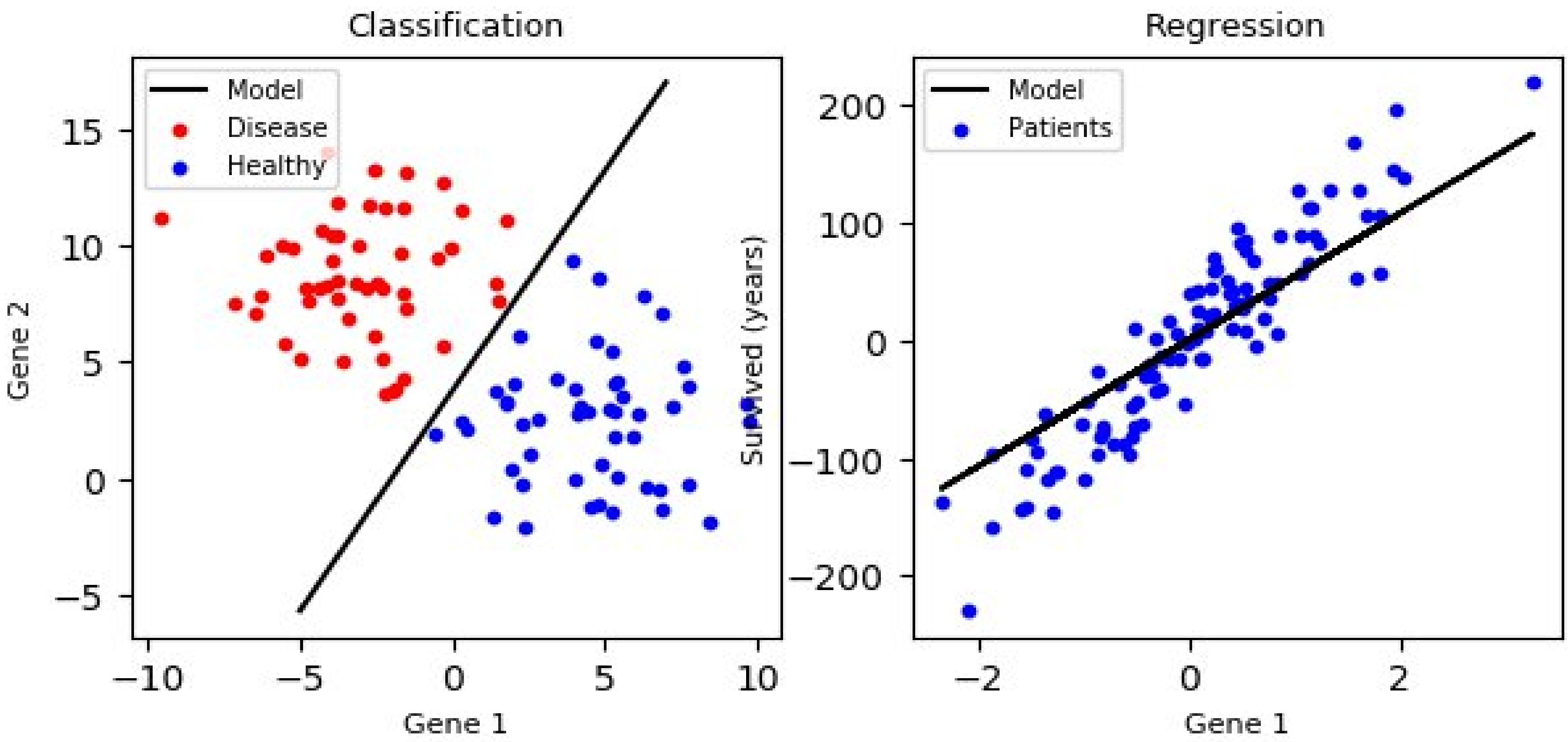
Classification



Regression

Supervised Learning - Regression vs Classification

Categorizing Supervised Learning Objectives



Simple Linear Regression & OLS

Foundations of Ordinary Least Squares (OLS) Optimization

- **The Linear Stochastic Model:** Representing the relationship as:

$$\hat{y}_i = \beta_0 + \beta_1 x_i$$

Minimizing the Cost Function: Solving the optimization problem by minimizing the Sum of Squared Residuals (SSR):

$$\min_{\beta_0,\beta_1}\left\{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2\right\} = \min_{\beta_0,\beta_1}\left\{\sum_{i=1}^{n}(y_i - [\beta_o + \beta_1\ x_i])^2\right\}$$

•**Geometric Interpretation:** Finding the hyperplane in a 2D space that minimizes the vertical displacement from the observed data points.

•**The Variance of the Error (ϵ):** Assuming independent and identically distributed (i.i.d.) residuals $(y_i - \hat{y}_i)$ to ensure BLUE: (Best Linear Unbiased Estimator) property.

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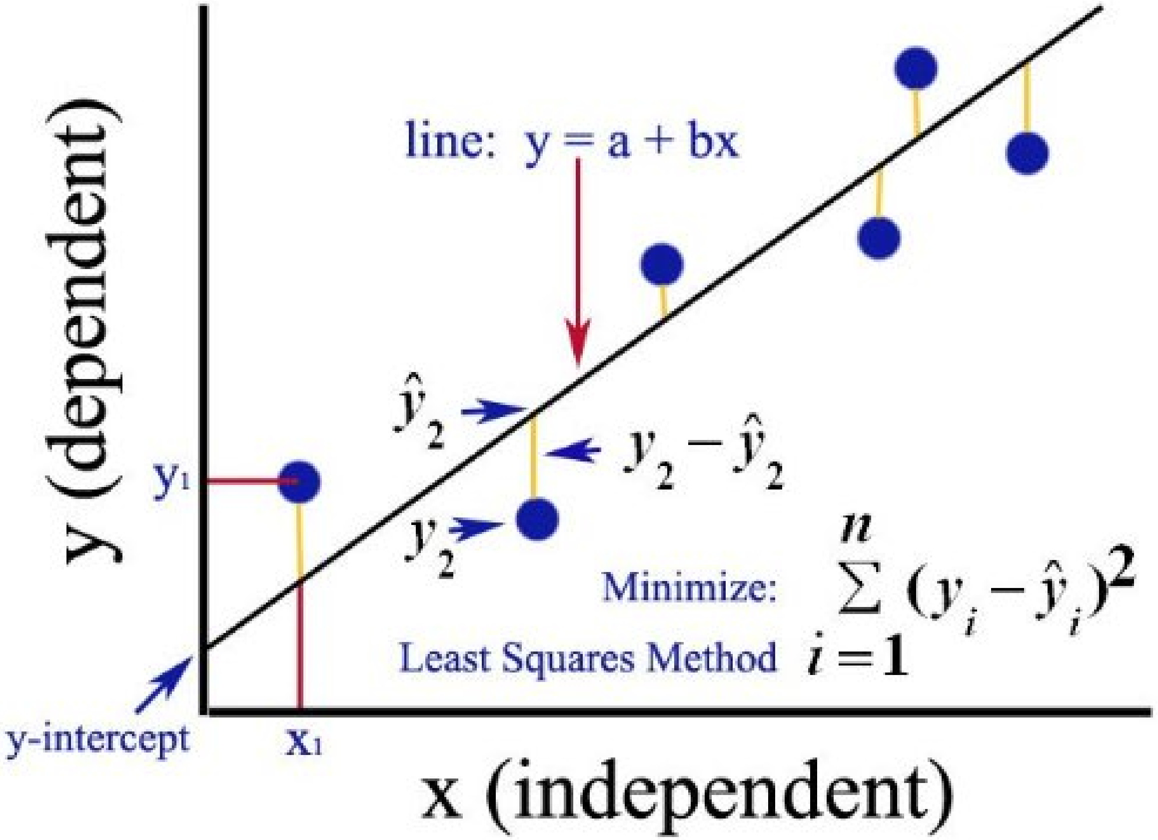
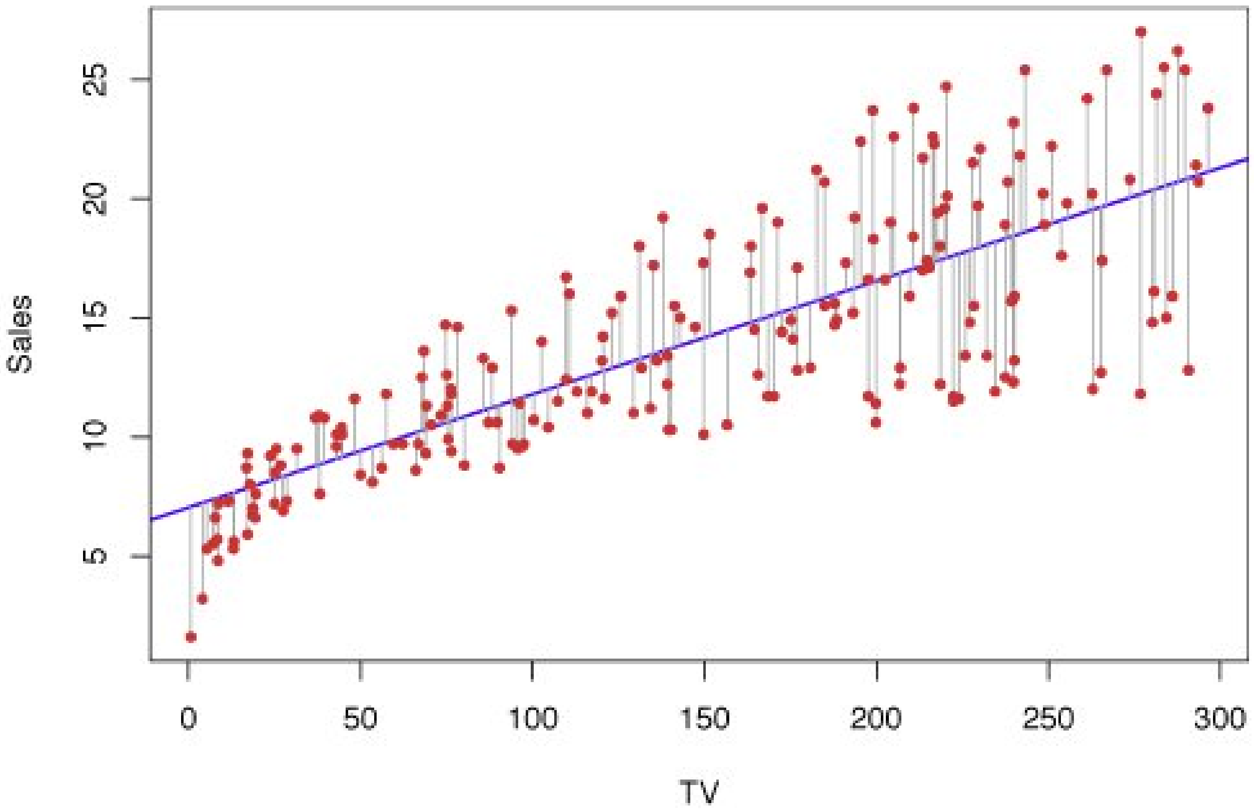




Model Performance & Error Metrics

Quantifying Precision in Operational Forecasting

• **The Variance Capture Ratio (R^2):** Measuring the percentage of system fluctuations explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2}, \quad R^2 \in [0,1]$$

•**Error Magnitude (MAE, MSE RMSE):** Quantifying prediction deviations in different ways.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \qquad MSE = \frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \quad RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

•**Relative Error Analysis (MAPE):** Normalizing performance metrics to allow cross-functional comparisons across different business scales and operational units.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n} \left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

Residual Diagnostics

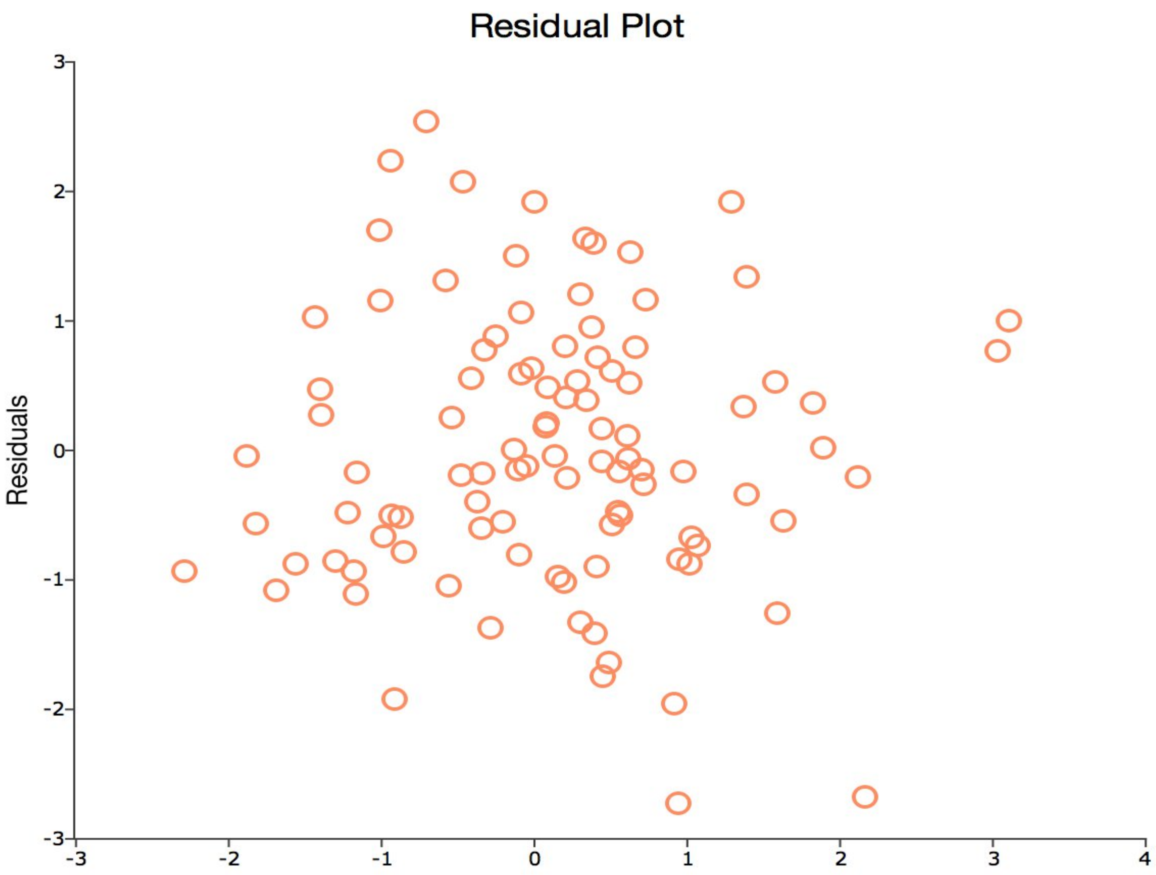
Validating Model Assumptions and Information Capture

- **Independence and Randomness:** Residuals $e_i = y_i - \hat{y}_i$ are random as white noise; any correlation or temporal pattern indicates a failure to capture systematic information.

- **Constant Variance (Homoscedasticity):** Verifying that the error spread remains uniform across the entire operational range, ensuring the model's reliability is consistent for both small and large scale predictions.

- **Normality of the Error Distribution:** Confirming that residuals follow a Normal distribution centered at zero $e_i \sim N\left(0, \sigma^2\right)$ which is essential for the statistical validity of p-values and confidence intervals.

- **Zero-Mean Assumption:** Validating that the model is unbiased on average, meaning the positive and negative deviations cancel out across the entire dataset.



The Overfitting Trap

Why a "Perfect" Model can be a Business Risk

- **Learning vs. Memorizing:** A common mistake is letting the model "memorize" historical anomalies instead of capturing the actual trend. This leads to a misleadingly high R^2.

- **Model Complexity vs. Degrees of Freedom:** Using a model that is too complex for the available data size leads to a high R^2 in the training set that vanishes in real-world applications.

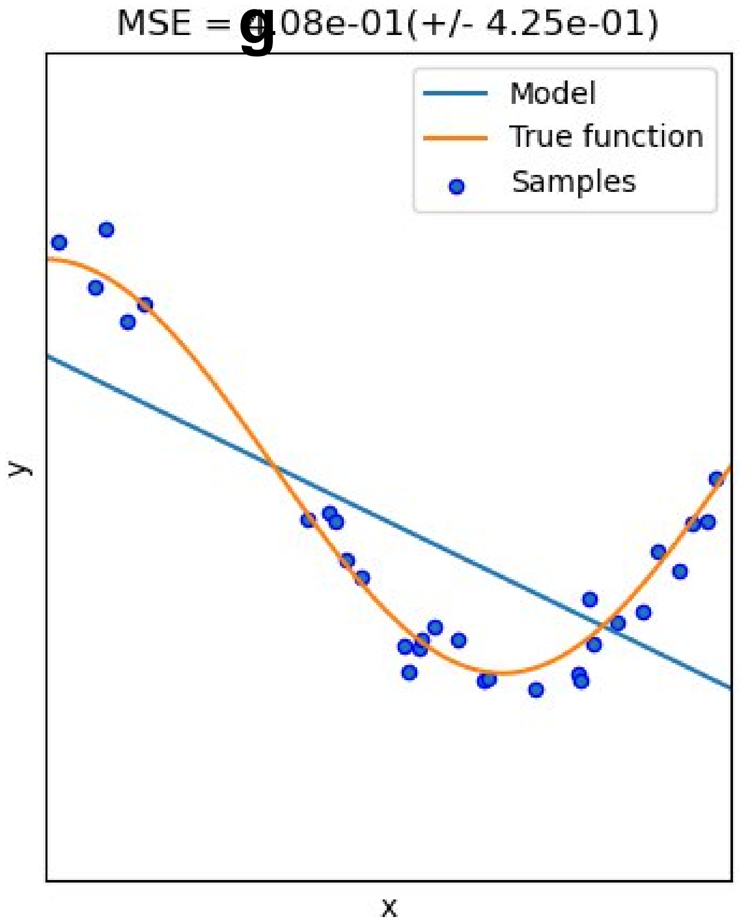
- **Testing with New Data:** The golden rule is to set aside a "Test" dataset. If the error spikes on data the model hasn't seen before, the model is not reliable for real-world deployment.

- **Simplicity over Extreme Precision:** A slightly less "accurate" but stable model is often better for decision-making than one that collapses when market conditions shift.

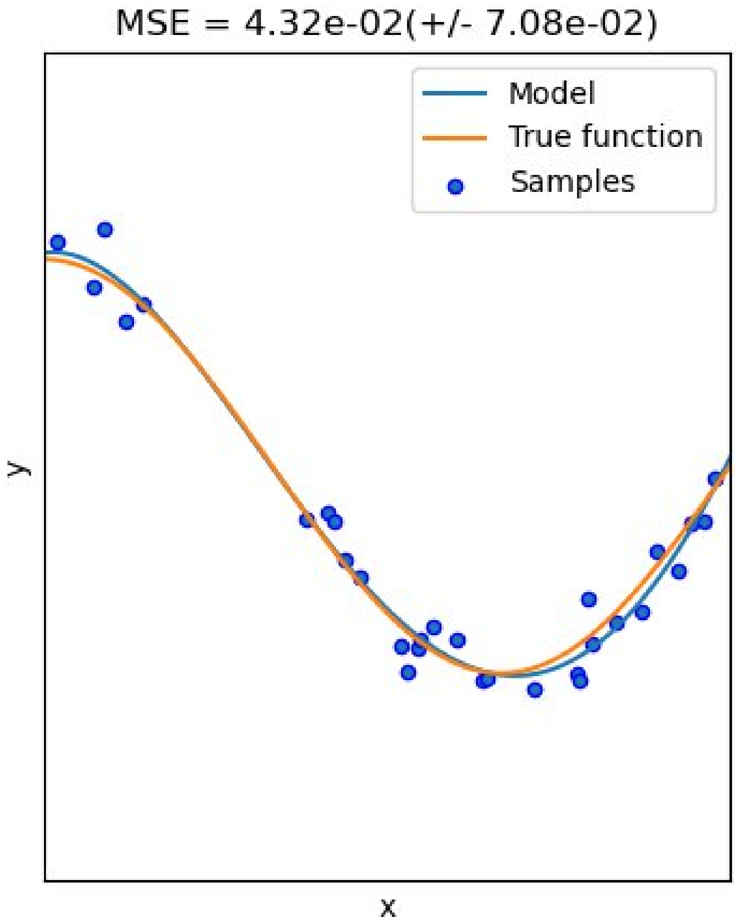
The Overfitting v/s Underfitting trade-off

Understanding your data before the modeling phase

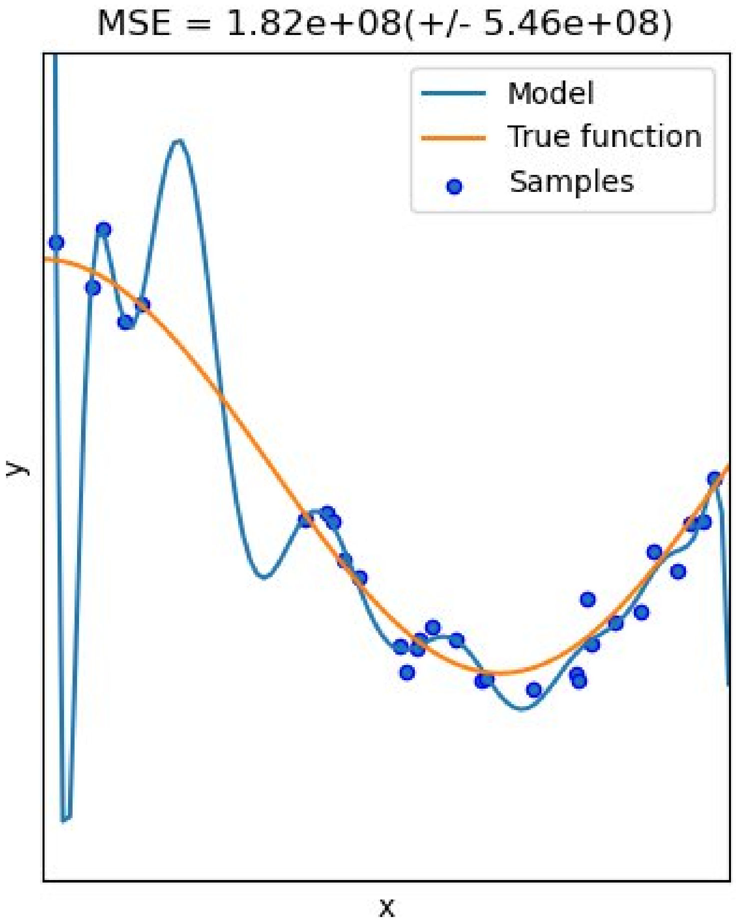
Underfitting



Reasonable fit

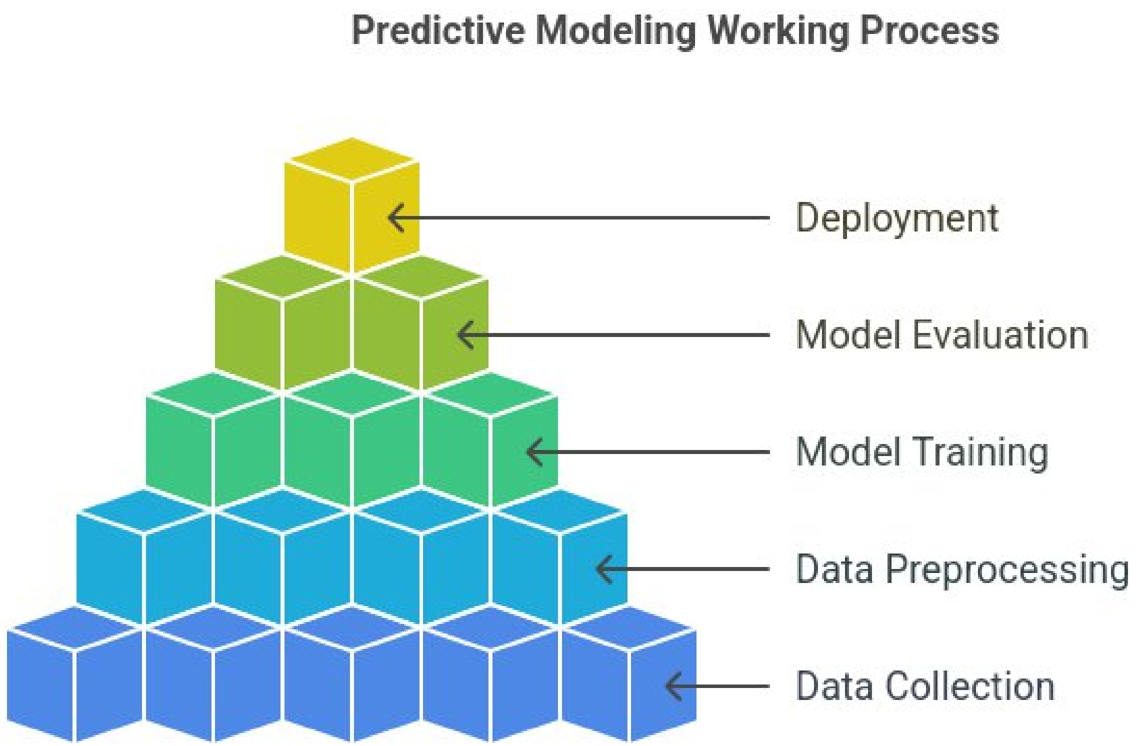


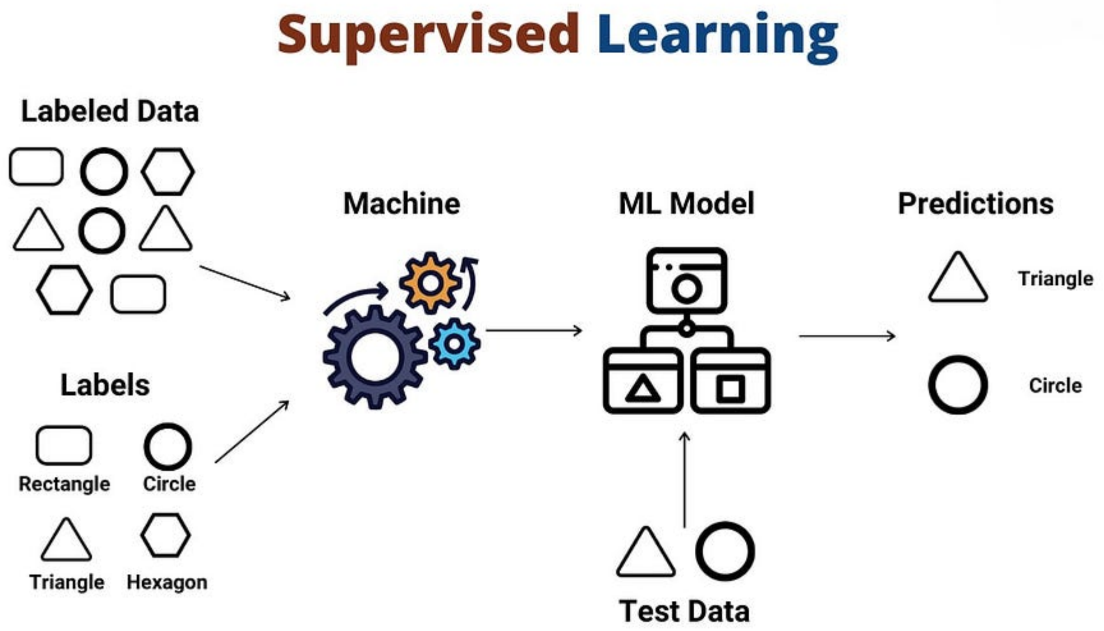
Overfitting



Data model decision triangle

The Strategic Framework for Predictive Business Intelligence





Practical Section

Google Colab — ML first project	
Regression model – Simple Linear regression model	
Housing dataset	
Forecast	
Performance metrics	

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